

Road Accident Pattern Clustering to Identify High-Risk Conditions

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• Problem Statement

Road accidents are influenced by multiple environmental and road-related factors such as weather conditions, road surface, lighting, and speed limits. Identifying patterns in these conditions can help understand high-risk scenarios.

This project aims to apply clustering techniques to group road accidents based on these factors and identify conditions that contribute to higher risk.

• Objective

- To analyze road accident data using clustering
 - To group accidents based on similar conditions
 - To identify high-risk accident patterns
 - To provide insights for safer road planning
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• Dataset Description

The dataset used is the **UK Road Safety Dataset**, specifically the file **Accident_Information.csv**.

It contains real-world accident data with features such as:

- Weather Conditions
- Road Surface Conditions
- Road Type
- Speed Limit
- Light Conditions
- Urban or Rural Area

- Time

The dataset contains a large number of records and represents real accident scenarios.

```
/tmp/ipykernel_24693/601556961.py:13: DtypeWarning: Columns (0) have mixed types. Specify dtype option on import or set low_memory=False.
df = pd.read_csv('Accident Information.csv')
Accident_Index 1st_Road_Class 1st_Road_Number 2nd_Road_Class \
0 200501BS00001 A 3218.0 NaN
1 200501BS00002 B 450.0 C
2 200501BS00003 C 0.0 NaN
3 200501BS00004 A 3220.0 NaN
4 200501BS00005 Unclassified 0.0 NaN

2nd_Road_Number Accident_Severity Carriageway_Hazards Date \
0 0.0 Serious NaN 2005-01-04
1 0.0 Slight NaN 2005-01-05
2 0.0 Slight NaN 2005-01-06
3 0.0 Slight NaN 2005-01-07
4 0.0 Slight NaN 2005-01-10

Day_of_Week Did_Police_Officer_Attend_Scene_of_Accident ... \
0 Tuesday 1.0 ...
1 Wednesday 1.0 ...
2 Thursday 1.0 ...
3 Friday 1.0 ...
4 Monday 1.0 ...

Police_Force Road_Surface_Conditions Road_Type \
0 Metropolitan Police Wet or damp Single carriageway
1 Metropolitan Police Dry Dual carriageway
2 Metropolitan Police Dry Single carriageway
3 Metropolitan Police Dry Single carriageway
4 Metropolitan Police Wet or damp Single carriageway
```

Fig1: df.head()

```
[5 rows x 34 columns]
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2047256 entries, 0 to 2047255
Data columns (total 34 columns):
#   Column                                                                                               Dtype
---  -
0   Accident_Index                                              object
1   1st_Road_Class                                              object
2   1st_Road_Number                                             float64
3   2nd_Road_Class                                              object
4   2nd_Road_Number                                             float64
5   Accident_Severity                                           object
6   Carriageway_Hazards                                         object
7   Date                                                        object
8   Day_of_Week                                                 object
9   Did_Police_Officer_Attend_Scene_of_Accident               float64
10  Junction_Control                                            object
11  Junction_Detail                                             object
12  Latitude                                                    float64
13  Light_Conditions                                            object
14  Local_Authority_(District)                                  object
15  Local_Authority_(Highway)                                  object
16  Location_Easting_OSGR                                       float64
17  Location_Northing_OSGR                                      float64
18  Longitude                                                    float64
19  LSOA_of_Accident_Location                                   object
20  Number_of_Casualties                                         int64
21  Number_of_Vehicles                                           int64
22  Pedestrian_Crossing-Human_Control                           float64
23  Pedestrian_Crossing-Physical_Facilities                    float64
```

Fig2: df.info()

• Data Preprocessing

The following preprocessing steps were performed:

- Selected relevant features for clustering
- Extracted **Hour** from the Time column
- Replaced invalid values like “*Unknown*” with NaN
- Removed missing values
- Converted categorical features into numerical format using One-Hot Encoding to avoid introducing ordinal relationships between categories.
- Standardized features using StandardScaler

```
11) 'Hour'
/ 3s ]]

# Replace invalid values with NaN
df_selected.replace(['Unknown', 'Not known', 'Data missing or out of range'], np.nan, inplace=True)

# Drop missing values
df_selected = df_selected.dropna()

# Check
print(df_selected.isnull().sum())

... /tmp/ipykernel_24693/826631444.py:4: UserWarning: Could not infer format, so each element will be parsed individually, falling back to 'dateutil'. To ensure parsing is consis
df['Hour'] = pd.to_datetime(df['Time'], errors='coerce').dt.hour
/tmp/ipykernel_24693/826631444.py:18: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user\_guide/indexing.html#returning-a-view-versus-a-copy
df_selected.replace(['Unknown', 'Not known', 'Data missing or out of range'], np.nan, inplace=True)
Weather_Conditions      0
Road_Surface_Conditions 0
Road_Type                0
Speed_limit             0
Light_Conditions         0
Urban_or_Rural_Area     0
Hour                    0
dtype: int64
```

Fig3: Missing values check output

• Algorithm Used

The **K-Means Clustering algorithm** was used.

Justification:

- Suitable for structured numerical data
- Efficient and easy to interpret
- Groups similar data points based on distance
- Helps identify patterns in accident conditions

The **Elbow Method** was used to determine the optimal number of clusters.

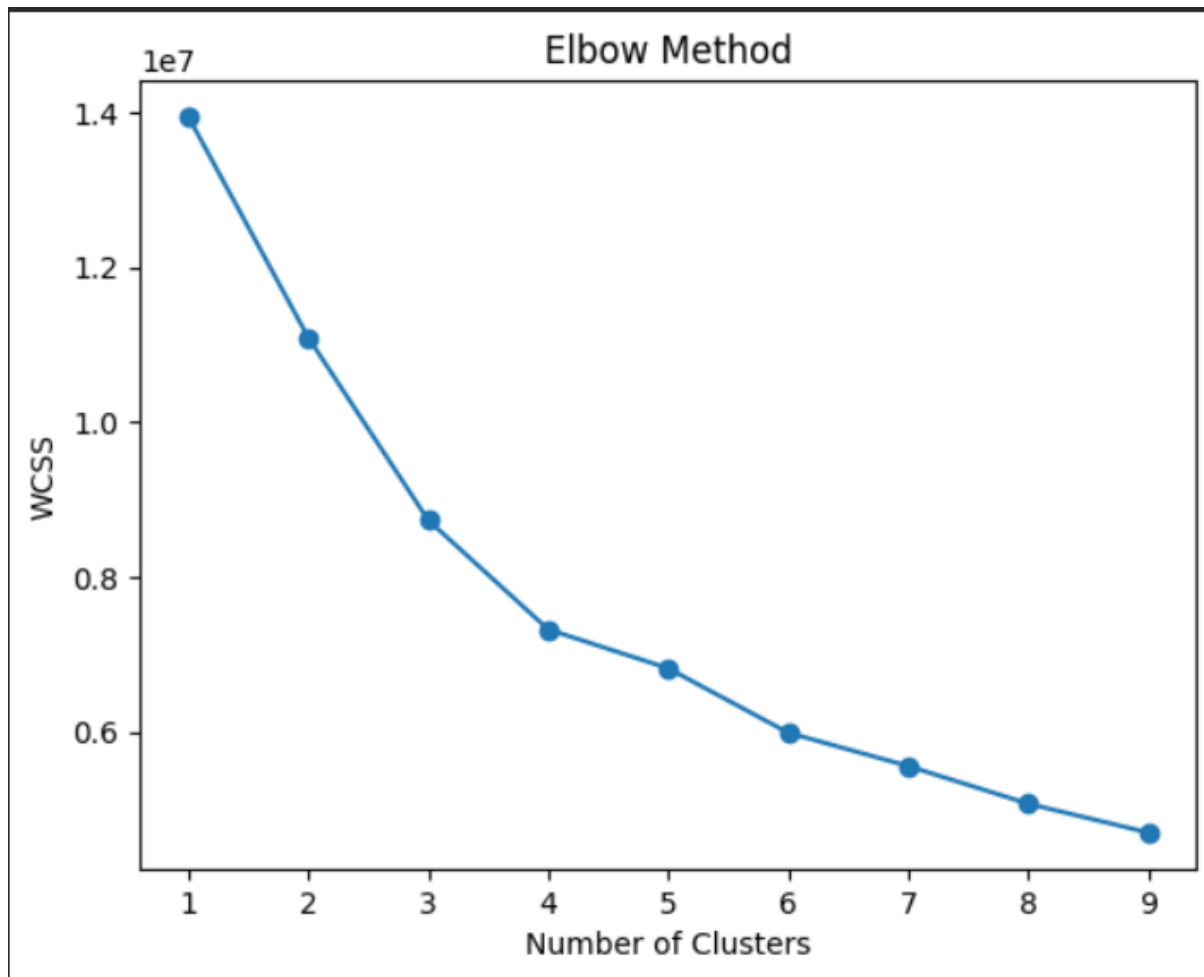


Fig4: Elbow graph

- From the graph, **K = 3** was selected.

• Methodology

1. Data loading and understanding
2. Data preprocessing and cleaning
3. Feature engineering (Hour extraction)
4. Encoding and scaling
5. Applying K-Means clustering
6. Visualization using PCA
7. Cluster interpretation

• Results / Output

The clustering results were visualized using:

- 2D raw feature plot
- PCA-based dimensionality reduction

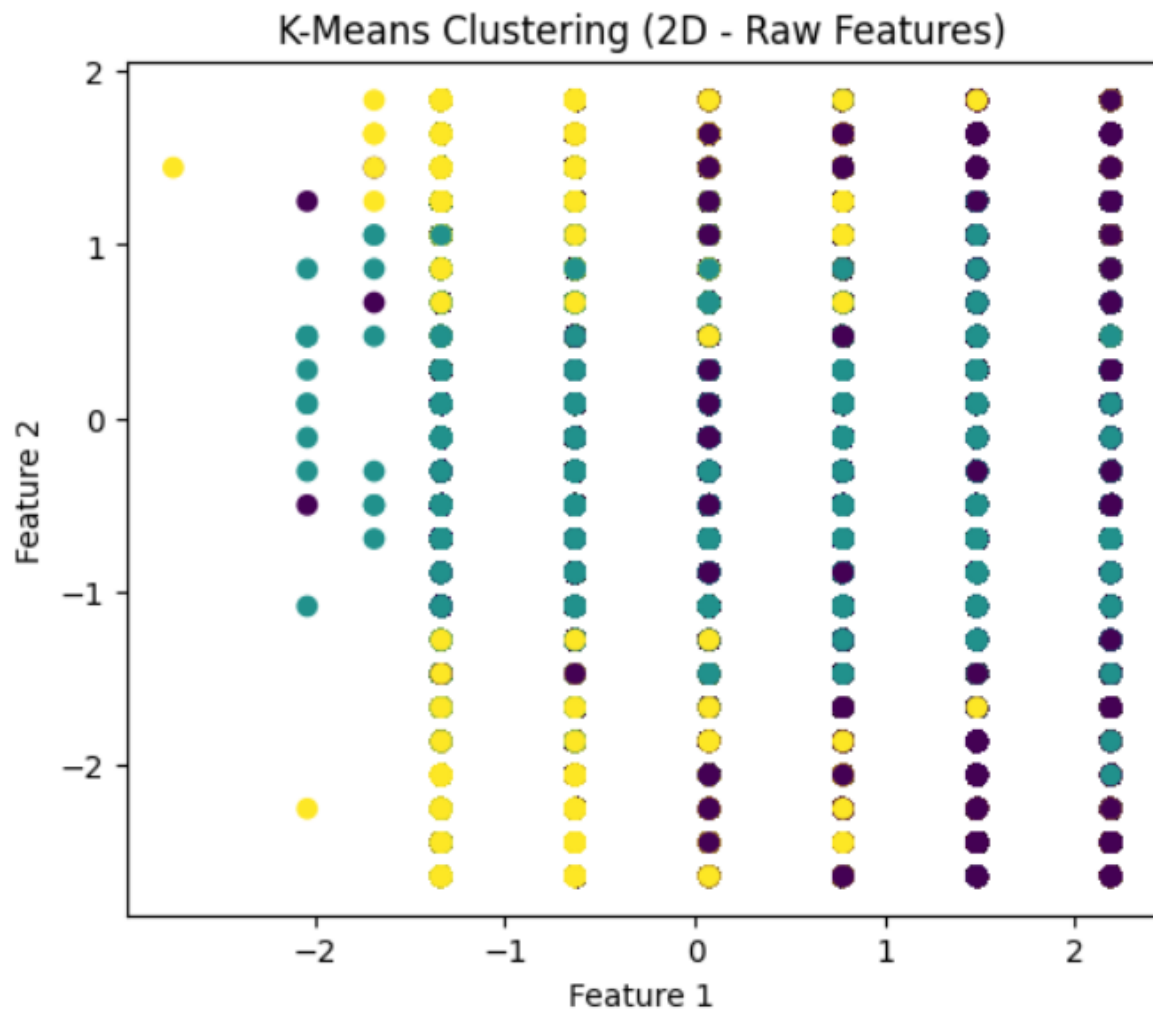


Fig5: 2D clustering plot

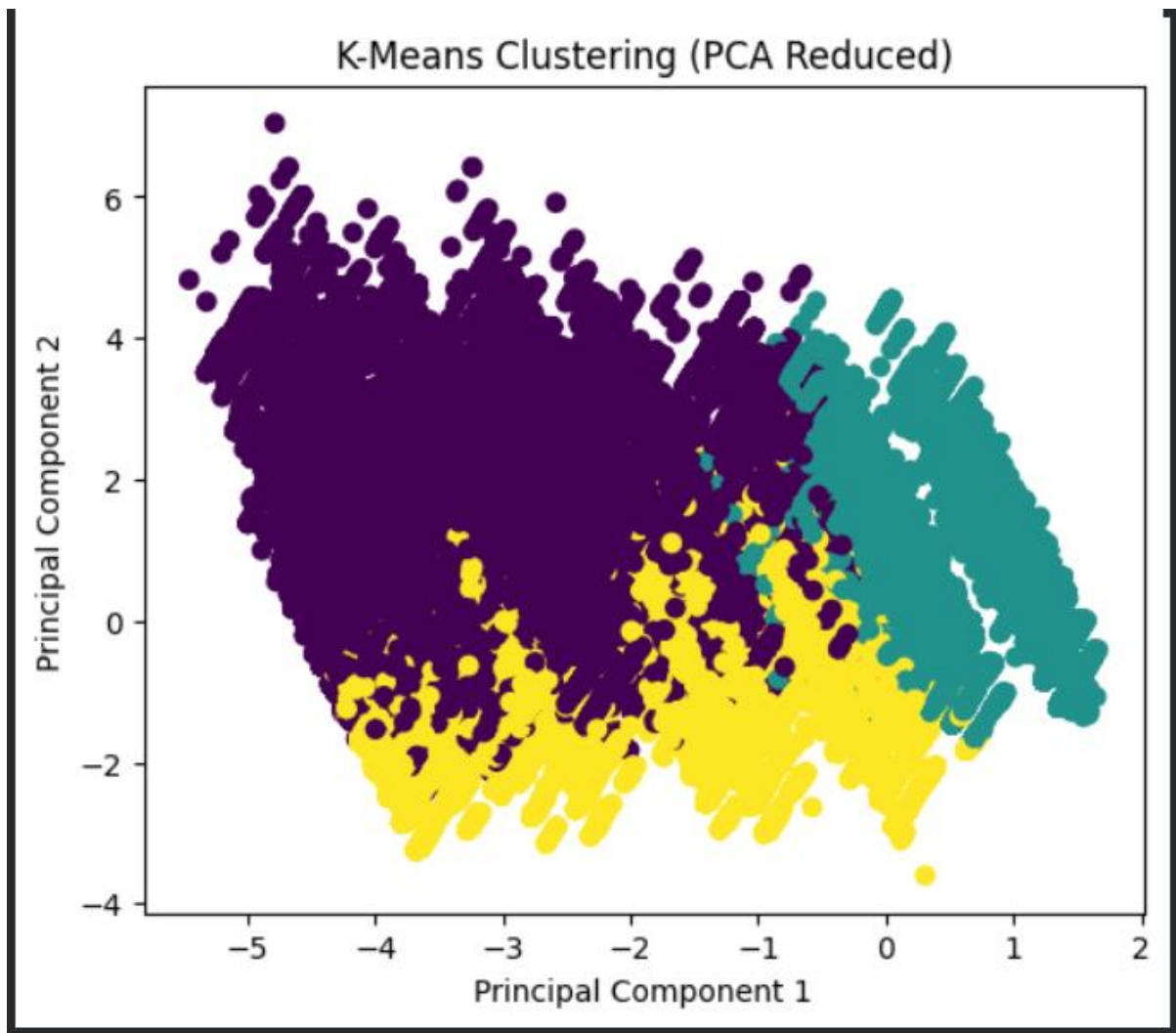


Fig6: PCA clustering plot

• Interpretation

Cluster analysis revealed three distinct accident patterns:

Cluster 0 — Adverse Weather & Wet Road Conditions

- Dominated by rainy weather and wet road surfaces
 - Includes both rural and urban areas, with rural slightly higher
 - Represents higher-risk conditions due to environmental factors
-

Cluster 1 — Normal Driving Conditions (Urban Daylight)

- Dominated by fine weather, dry roads, and daylight
 - Mostly urban areas
 - Represents normal driving conditions with lower risk
-

Cluster 2 — Low Visibility / Night Driving Conditions

- Dominated by darkness and reduced visibility
 - Mostly urban areas
 - Represents conditions where visibility is limited, increasing accident severity
-

◆ Weather_Conditions distribution per cluster:

Cluster	Weather_Conditions	
0	Raining no high winds	179305
	Fine no high winds	76148
	Other	32743
1	Fine no high winds	1254039
	Fine + high winds	10405
2	Fine no high winds	299563
	Raining no high winds	58616
	Other	10774

Name: count, dtype: int64

◆ Road_Surface_Conditions distribution per cluster:

Cluster	Road_Surface_Conditions	
0	Wet or damp	255001
	Dry	57004
	Frost or ice	16555
1	Dry	1093662
	Wet or damp	154782
	Frost or ice	14572
2	Dry	227449
	Wet or damp	149496
	Frost or ice	8478

Name: count, dtype: int64

```

♦ Road_Type distribution per cluster:
...
Cluster Road_Type
0 Single carriageway 253359
  Dual carriageway 63953
  Roundabout 16035
1 Single carriageway 959346
  Dual carriageway 178197
  Roundabout 86082
2 Single carriageway 284433
  Dual carriageway 56026
  Roundabout 31143
Name: count, dtype: int64

♦ Light_Conditions distribution per cluster:

Cluster Light_Conditions
0 Daylight 206552
  Darkness - no lighting 110698
  Darkness - lights lit 17517
1 Daylight 1250496
  Darkness - lighting unknown 13752
  Darkness - lights unlit 132
2 Darkness - lights lit 377602
  Darkness - lights unlit 6754
  Darkness - lighting unknown 1410
Name: count, dtype: int64

```

♦ Urban_or_Rural_Area distribution per cluster:

```

Cluster Urban_or_Rural_Area
0 Rural 211569
  Urban 129935
  Unallocated 36
1 Urban 827488
  Rural 436861
  Unallocated 95
2 Urban 324669
  Rural 61126
  Unallocated 13

```

Name: count, dtype: int64

```

Cluster Cluster_Label
0 0 Adverse Weather & Wet Road Conditions
1 2 Low Visibility / Night Driving Conditions
2 2 Low Visibility / Night Driving Conditions
3 1 Normal Driving Conditions (Urban Daylight)
4 2 Low Visibility / Night Driving Conditions

```

Fig7: Cluster summary table

• Additional Analysis

Cluster distribution shows that most accidents occur under normal urban daytime conditions. This is expected due to higher traffic exposure rather than higher risk.

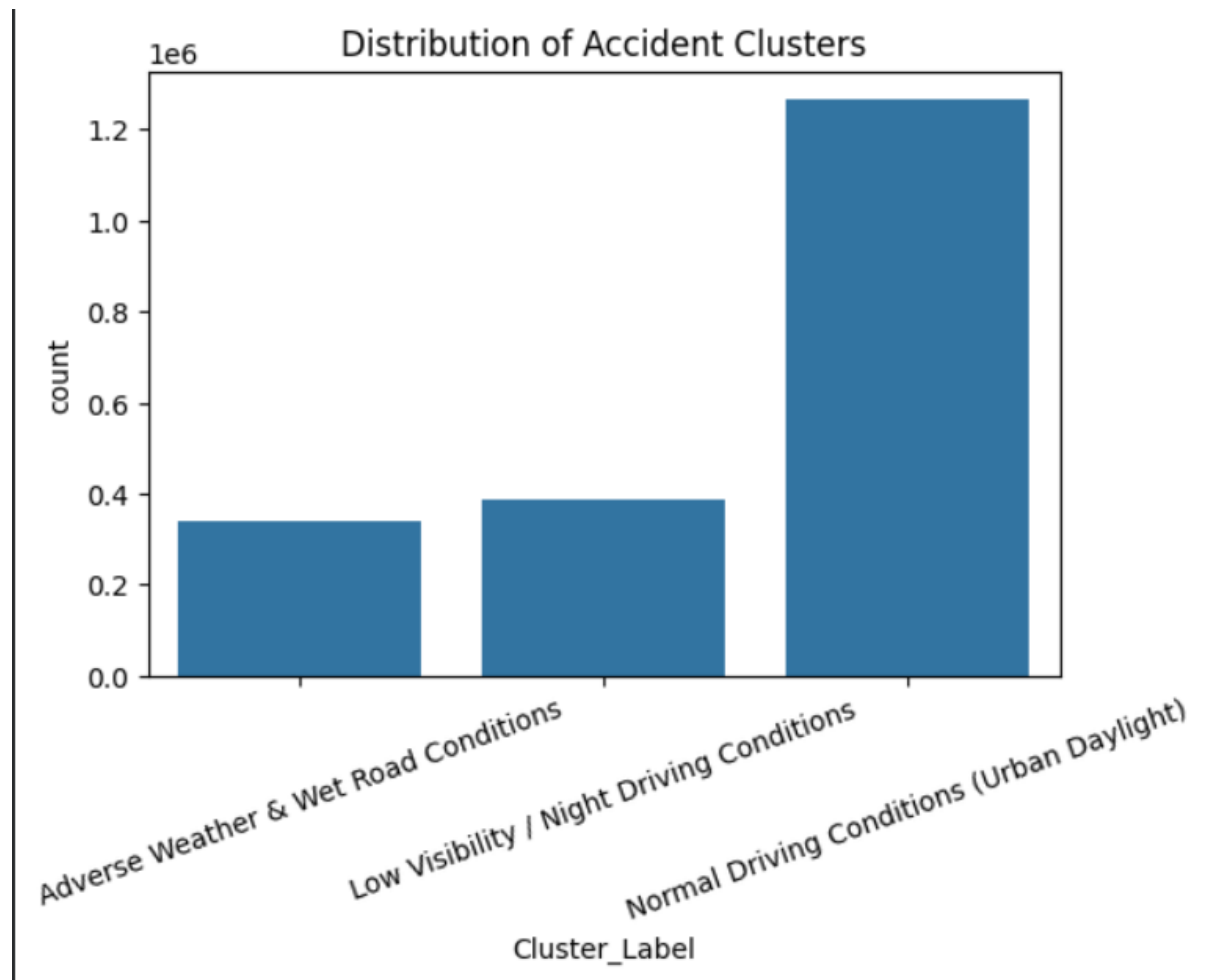


Fig8: Cluster distribution plot

Severity analysis indicates that:

- Adverse weather conditions have the highest fatality rate (~2.05%), making them the most dangerous
- Night-time conditions have the highest proportion of serious injuries (~15.38%) due to reduced visibility
- Urban daytime conditions have the lowest fatality rate (~1.08%), making them the safest despite having the highest number of total accidents

This highlights the importance of analysing percentages rather than absolute counts when comparing clusters of different sizes.

◆ Severity Distribution (Counts):

Accident_Severity Cluster_Label	Fatal	Serious	Slight
Adverse Weather & Wet Road Conditions	7023	48467	286050
Low Visibility / Night Driving Conditions	5224	59352	321232
Normal Driving Conditions (Urban Daylight)	13659	172840	1077945

◆ Severity Distribution (Percentage):

Accident_Severity Cluster_Label	Fatal	Serious	Slight
Adverse Weather & Wet Road Conditions	0.020563	0.141907	0.837530
Low Visibility / Night Driving Conditions	0.013540	0.153838	0.832621
Normal Driving Conditions (Urban Daylight)	0.010802	0.136692	0.852505

Fig9: Severity analysis table

• Conclusion

This project applied K-Means clustering to analyze road accident patterns based on environmental and road-related factors such as weather conditions, road surface, lighting, location type, and time of occurrence.

The clustering results revealed three meaningful groups of accident scenarios:

- **Adverse Weather & Wet Road Conditions** showed the **highest fatality rate (≈2.05%)**, indicating that poor weather and slippery roads significantly increase accident severity.
- **Low Visibility / Night Driving Conditions** exhibited the **highest proportion of serious injuries (≈15.38%)**, suggesting that reduced visibility contributes to more severe but not always fatal accidents.
- **Normal Driving Conditions (Urban Daylight)** had the **lowest fatality rate (≈1.08%)**, making it the safest cluster despite having the highest number of total accidents.

It is important to note that although the favourable conditions cluster recorded the highest number of fatal accidents, this is due to its significantly larger size. When normalized using percentages, it is actually the least risky environment per accident. This highlights that accident frequency is influenced by exposure (higher traffic in

normal conditions), whereas accident severity is more influenced by environmental risk factors.

Overall, the analysis demonstrates that accident risk is not evenly distributed and depends on multiple interacting factors such as weather, lighting, and road conditions. Clustering helped uncover these hidden patterns, providing insights that can support road safety improvements, such as better infrastructure in rural and wet conditions, and enhanced visibility measures for night-time driving.

References

- UK Road Safety Dataset
- Scikit-learn documentation
- Course material